WORKED SOLUTION

STEP II 2004, Question 13

Probability and Statistics • Optimal Stopping

STEP II 2004, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q13

The key idea

Think one draw at a time. If she has safely collected n white balls, one more draw either adds £1 or wipes out the £n she is holding, so its expected effect is easy to write down. The total expected reward for a target of N balls goes up exactly when the expected gain of the extra draw is positive, so the best target is the last one at which that gain is still non-negative.

Solution

Throughout we assume $1 \le r < b$, so there is at least one red and at least one white ball. (With $r = 0$ she simply takes every ball.)

The expected gain on the next draw

Suppose her first n draws were all white. The bag now holds $b - n$ balls: r red and $b - n - r$ white, each equally likely to come out next (we need $n \le b - r - 1$ for there to be a white ball left; in any case $b - n \ge r \ge 1$).

- With probability $\dfrac{b-n-r}{b-n}$ the ball is white, and her reward goes from £n to £$(n+1)$: a gain of $+1$.
- With probability $\dfrac{r}{b-n}$ the ball is red, and she loses the £n she would otherwise have had: a gain of $-n$.

So her expected gain from the next draw is

$$G_n = \frac{(b-n-r)\cdot 1 + r\cdot(-n)}{b-n} = \frac{b-r-n(r+1)}{b-n}.$$

The denominator is positive, so $G_n = 0$ exactly when $b - r = n(r+1)$, that is when

$$n = \frac{b-r}{r+1},$$

as required. More usefully, $G_n > 0$ when $n < \dfrac{b-r}{r+1}$ and $G_n < 0$ when $n > \dfrac{b-r}{r+1}$.

The best number of balls

Let P_N be the probability that the first N draws are all white, and let E_N be her expected total reward if she aims to remove N balls. She gets £N if those N balls are all white and nothing otherwise, so

$$E_N = NP_N.$$

Ask yourself How does the expected gain on one draw relate to the total expected reward E_N?

Compare aiming for N with aiming for $N+1$. The two plans differ only if the first N balls are all white (otherwise she has already lost under both), which happens with probability P_N; and in that case the extra draw changes her reward by an amount with expected value G_N. Hence

$$E_{N+1} - E_N = P_N\, G_N = P_N\, \frac{b-r-N(r+1)}{b-N}.$$

(One can also check this directly from $P_{N+1} = P_N \cdot \frac{b-r-N}{b-N}$.)

For $N \le b-r$ we have $P_N > 0$, so $E_{N+1} - E_N$ has the sign of $b-r-N(r+1)$. Writing $k = N+1$, this is the sign of

$$(b+1) - k(r+1).$$

Put $m = \dfrac{b+1}{r+1}$. Then $E_k > E_{k-1}$ when $k < m$, $E_k = E_{k-1}$ when $k = m$, and $E_k < E_{k-1}$ when $k > m$. So the sequence $E_0, E_1, E_2, \ldots$ increases strictly up to $E_{\lfloor m \rfloor}$ (with a tie $E_{m-1} = E_m$ if m is an integer) and then decreases strictly, at least while $k - 1 \le b - r$. Beyond that $P_N = 0$ and $E_N = 0$, which is no better.

We should check that $\lfloor m \rfloor$ is a sensible target, i.e. $\lfloor m \rfloor \le b-r$. If $b = r+1$ then $m = \frac{r+2}{r+1}$, so $\lfloor m \rfloor = 1 = b-r$. If $b \ge r+2$ then

$$(b-r)(r+1) - (b+1) = r(b-r-1) - 1 \ge r - 1 \ge 0,$$

so $m \le b-r$. Either way the maximum of E_N is attained at

$$n = \left\lfloor \frac{b+1}{r+1} \right\rfloor,$$

the integer part of $\frac{b+1}{r+1}$, as required.

This target is also the best she can do with any stopping rule: the only thing she ever learns before deciding whether to carry on is that every ball so far was white (if not, the game is over), so every rule amounts to choosing a number N in advance.

The case $r = 1$

With one red ball, the first N draws avoid it exactly when the red ball is in one of the last $b - N$ positions, all positions being equally likely. So

$$P_N = \frac{b-N}{b}, \qquad E_N = \frac{N(b-N)}{b}.$$

The best target is $n = \left\lfloor \frac{b+1}{2} \right\rfloor$.

b even. Then $\frac{b+1}{2} = \frac{b}{2} + \frac{1}{2}$, so $n = \frac{b}{2}$ and

$$E_n = \frac{\frac{b}{2} \cdot \frac{b}{2}}{b} = \frac{b}{4}.$$

Her expected total reward is £$\frac{1}{4}b$, as required.

b odd. Then $n = \frac{b+1}{2}$, so $b - n = \frac{b-1}{2}$ and

$$E_n = \frac{\frac{b+1}{2} \cdot \frac{b-1}{2}}{b} = \frac{b^2-1}{4b}.$$

Her expected total reward is £$\frac{b^2-1}{4b}$. (Here $m = \frac{b+1}{2}$ is an integer, so aiming for $\frac{b-1}{2}$ balls gives the same value, as the symmetric formula $N(b-N)/b$ shows.)

Checks

- For $r = 1$, $E_N = N(b-N)/b$ is a downward parabola in N with vertex at $N = b/2$, which matches both answers: exactly $b/4$ at the vertex when b is even, and slightly less, $\frac{b}{4} - \frac{1}{4b}$, when b is odd.
- Take $b = 7$, $r = 2$: the formula gives $n = \lfloor 8/3 \rfloor = 2$, and direct computation gives $E_0, \ldots, E_5 = 0,\ \frac{5}{7},\ \frac{20}{21},\ \frac{6}{7},\ \frac{4}{7},\ \frac{5}{21}$, peaking at $N = 2$. We confirmed this by enumerating all 7! orderings of the balls, and checked the formula for n against a full optimal-stopping calculation for every $b \le 24$.
- For $b = 9$, $r = 1$ the answer is $\frac{80}{36} = \frac{20}{9} \approx 2.22$, just under $\frac{9}{4}$.

Common slips

- Forgetting that a red ball costs her the £n already collected, and writing the expected gain as just $\frac{b-n-r}{b-n}$.
- Confusing "n balls already drawn" with "target of n balls": the gain G_n is zero at $n = \frac{b-r}{r+1}$, which is why the best *target* is one more, $\frac{b-r}{r+1} + 1 = \frac{b+1}{r+1}$.
- Missing the tie when $\frac{b+1}{r+1}$ is an integer: then both n and $n-1$ are optimal.

Beyond STEP

This is a small *optimal stopping* problem, and the argument is the "one-step look-ahead" rule: stop as soon as one more step has negative expected gain. That rule is optimal whenever the problem is *monotone*, meaning that once continuing looks bad it stays bad, as here, where G_n decreases in n. The same ideas lead to the secretary problem and to martingale methods in first-year and second-year probability. Grimmett and Welsh, *Probability: An Introduction*, is a good place to build the conditional-expectation tools behind it.

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